Understanding Transformer Architecture

Transformers process language by ingesting entire sentences simultaneously rather than word-by-word. Three core components cooperate to enable this deep context modeling: Vector Embeddings, the Self-Attention Mechanism, and Position-Wise Feed-Forward Neural Networks.

1. Vector Embedding: Giving Words a Numerical Identity

Computers require numbers rather than text to calculate relations. Vector embedding converts a word into a high-dimensional mathematical coordinate that captures its real-world semantic meaning.

The Semantic Space: Words with similar usage profiles or meanings are mapped close together in this multi-dimensional space. The vectors for *'king'* and *'queen'* reside near each other, while *'banana'* resides far away.

Adding Position: Since Transformers evaluate inputs concurrently, they do not implicitly track word order. To resolve this, a **Positional Encoding** vector is added directly to the word embedding, ensuring the model distinguishes structure.

2. Self-Attention Mechanism: Calculating Context and Relationships

Self-attention enables the model to dynamically score which other words in a sentence are most relevant to understanding the target word. For example, in *'The animal didn't cross the street because it was too tired'*, it computes that **'it'** strongly correlates to **'animal'**.

The system relies on a Query, Key, and Value (QKV) mathematical projection:

Query (Q): What the current word token is searching for or trying to resolve.

Key (K): The identity label of other words in the sentence used for matching against queries.

Value (V): The actual informational content passed along once an attention match is scored.

The Calculation Steps: First, the model multiplies the *Query* of a word with the *Keys* of all words to compute attention weights. Second, a softmax function normalizes these weights. Third, the normalized weights are multiplied by the *Values* to amplify relevant attributes and suppress noise.

Multi-Head Attention: This process runs across multiple parallel 'heads'. One head may capture syntax (nouns to verbs), while another tracks pronouns or factual dependencies.

3. Position-Wise Feed-Forward Neural Network (FFNN)

After gathering context via attention, the representations must undergo non-linear transformations to build robust feature extractions. This task falls to the Feed-Forward Neural Network.

Isolated Processing: Every word vector passes through the exact same FFNN layer completely independently, localized to its own position.

Non-Linear Transformation: By applying activation functions like ReLU or GELU, the network captures complex, high-level abstract interactions that simple linear operations cannot map.

Synthesis: While self-attention is dedicated to *gathering context* from external tokens, the feed-forward network is built to *consolidate and process* that context, preparing the vectors for the subsequent layer.

Component	Core Responsibility	Real-World Analogy
Vector Embedding	Converts text into mathematical coordinates based on definition and word order.	Looking up baseline definitions in a specialized multi-dimensional dictionary.
Self-Attention	Calculates relational weights to adapt word meanings based on surrounding text context.	Disambiguating whether 'bank' refers to a river side or a financial institution.
Feed-Forward Network	Applies non-linear transformations to synthesize gathered context into deep feature representations.	Synthesizing all accumulated clues to form a finalized, concrete understanding.